

WASP-2b

R-1648

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-2_160731 · created 2026-08-02T07:55:34+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457600.7995 +/- 0.0044 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.149 +/- 0.044
Transit depth (Rp/Rs)^2	2.23 +/- 1.32 [%]
Orbital Inclination (inc)	84.0 +/- 2.74
Airmass coefficient 1 (a1)	1.098 +/- 0.053
Airmass coefficient 2 (a2)	-0.08 +/- 0.11
Scatter in the residuals of the lightcurve fit is	4.8 %
Best Comparison Star	#3 - [105, 456]
Optimal Method	PSF photometry
Transit Duration (day)	0.054 +/- 0.036

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.149 $\pm$ 0.044 vs 0.1326 (deviation 0.25 σ)
Duration fit vs published	0.054 d vs 0.0737 d (deviation 0.37 σ)
Mid-time O-C	11.3 min
Depth SNR	2.3
Residual scatter	4.8 %

Light curve

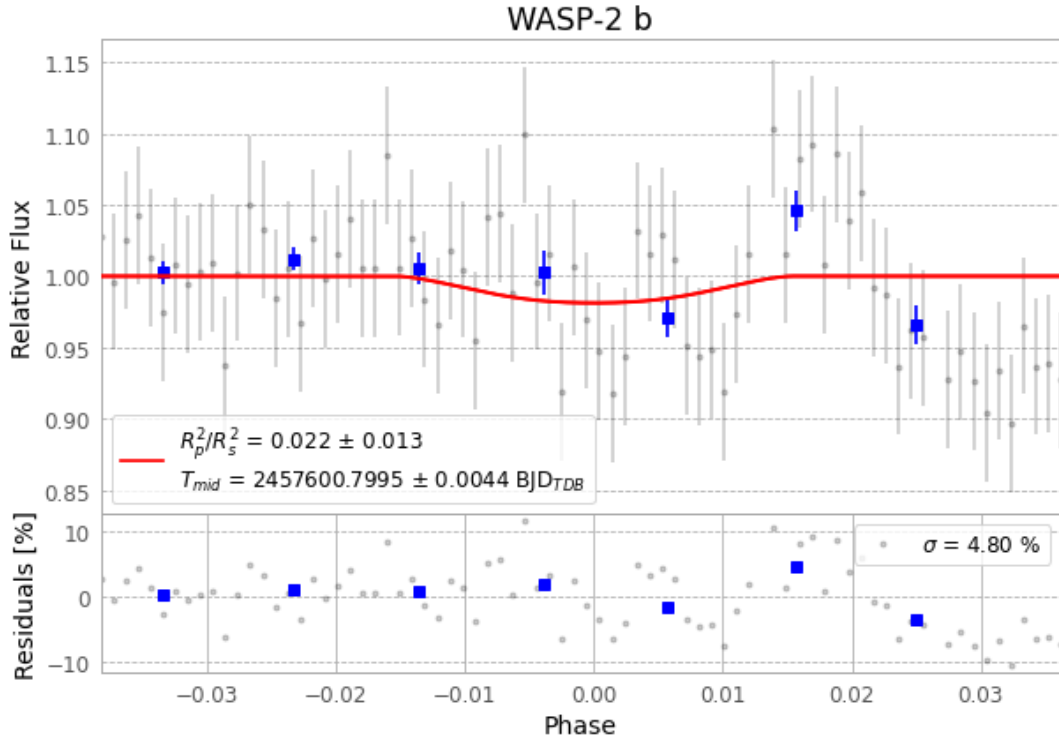


Figure 1 — FinalLightCurve_WASP-2 b_2016-07-30.png (SHA-256 2511ab54f728bf1e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [224, 16], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 cfbae309e5ec349d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

78 FITS frames (51 MB), WASP-2160731050313.FITS → WASP-2160731085412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-160731.log (63c20f91a911...), FinalLightCurve_WASP-2 b_2016-07-30.png (2511ab54f728...), FinalLightCurve_WASP-2 b_2016-07-30.csv (56952af782f9...)

Compute resources

Wall time 751.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 07:55Z.